

Tidewater Pay

A demo mobile-style wallet app with facial biometric login. Single file, no build step.

Demo only. Face data, PINs and balances are stored in your browser (localStorage). Do not use it for real money or real accounts.

Features

- Create a wallet with a username, a 6-digit backup PIN and a 5-pose face enrollment
- Sign in with your face (matches your enrolled face and requires a blink, so a still photo is rejected)
- PIN fallback after 3 failed face attempts; 5 wrong PINs lock sign-in for 30 seconds
- Wallet and savings balances, activity list, hide/show balance
- Send to a bank account or move money between wallet and savings
- Transfers of ₦50,000 or more to a bank need a fresh face check
- Re-scan or remove your face data from the Me tab

Run locally

The camera only works on `localhost` or `https` , so don't open the file directly.

Windows:

```
py -m http.server 8000
```

Then open <http://localhost:8000>. You need internet on first load for the fonts, the face-api.js library and its models.

Publish with GitHub Pages

1. Push this folder to a GitHub repository.
2. Go to **Settings** → **Pages**.
3. Under **Build and deployment**, choose **Deploy from a branch**, pick `main` and /

(root) , then save.

4. Open the https link GitHub gives you. The camera works there, including on your phone.

Settings you can change

Near the top of the script in `index.html` :

Name	What it does
<code>MATCH_THRESHOLD</code>	Face match strictness. Lower is stricter (default 0.5).
<code>STEP_UP_KOBO</code>	Transfer size that triggers a face check (default ₦50,000).
<code>MAX_FACE_FAILS</code>	Failed face attempts before PIN is required (default 3).
<code>BANKS</code>	Banks shown in the send form.

Limits

- Matching happens in the browser, so a determined attacker could bypass it. A real app must match faces on a server.
- The blink check is basic. Real apps use certified liveness detection.
- No real payments, accounts or bank connections.

Credits

Face detection and recognition by [@vladmandic/face-api](https://github.com/vladmandic/face-api).